

Ahmad Khaled Elshafee

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SUMMARY

Mechatronics Engineer and AI-driven robotics researcher specializing in embedded systems, biomedical devices, and intelligent healthcare solutions. Proven experience leading multidisciplinary international projects, including mega AI conferences and government tech initiatives. Seeking an impactful role in Embedded Systems, Medical Robotics, or AI-driven healthcare systems to develop intelligent, scalable, and highly accurate solutions.

EDUCATION

Master's Degree in Medical Robotics & AI in Healthcare (In Progress)

Mansoura University, Egypt

Bachelor of Mechatronics Engineering | Class of 2023

Horus University, Egypt

- **GPA:** 3.15
- **Graduation Project:** Dr. HEMA – Medical Diagnostic Robot

EXPERIENCE

Software Engineer | HudHud Project (Saudi Arabia)

Nov 2023 – May 2024

- Spearheaded cross-functional team integration and software development for the national HudHud initiative.
- Promoted to Vice Head of Development in March 2024 after successfully aligning multiple technical teams and streamlining the software delivery lifecycle.

Expert Trainer & Project Manager | BambooGeeks

Dec 2021 – May 2024

- **International Event Management:** Traveled to Tunisia in cooperation with the Tunisian Ministry of Industry to manage and execute "Tunisia AI Hack," a mega AI conference.
- **Remote Tech Leadership:** Managed and organized the Saudi Tech LEAP 2024 remotely, coordinating with international stakeholders.
- **Technical Mentorship:** Managed the Flutter and Android Academies, supervising multiple technical projects and mentoring students in software and system development.

KEY PROJECTS

Dr. HEMA – AI-Powered Medical Diagnostic Robot | Aug 2022 – Jul 2023

- **Leadership & Design:** Led a multidisciplinary team of engineers and scientists to develop a machine learning-powered robot aimed at providing basic healthcare to underserved communities.
- **Technical Implementation:** Integrated biomedical sensors to collect vital signs, blood pressure, and blood sugar levels. Architected ML models to identify patterns and diagnose common diseases.
- **Clinical Success:** Conducted clinical trials to evaluate efficacy, achieving a diagnostic accuracy of **>90%**. Designed the system to be highly affordable and operable by individuals with limited medical training.

Smart University System | Mar 2021 – Jul 2022

- **IoT & Embedded Systems:** Engineered an IoT infrastructure utilizing ESP boards for live bus tracking (speed, location) and a streamlined campus RFID payment system.
- **Mobile App Integration:** Deployed an application featuring dynamic timetables, real-time transportation tracking, attendance via facial recognition, and an automated student complaint escalation system.

Smart Hand Sanitizer & Health Monitor | Jan 2021 – Mar 2021

- **Hardware Engineering:** Built an ESP32-controlled COVID-19 safety device featuring a 7-inch TFT screen, integrating a MAX30102 sensor (HR/O2), an MLX90614 IR thermometer (via I2C), and a Sharp IR sensor.
- **System Integration:** Incorporated RFID technology to automate university lecture attendance while simultaneously verifying and logging student health metrics.

Additional Robotics & Software Projects

- **Robotics:** EV-ion Robot, InMoov Humanoid Robot
- **Software:** Dr. DERMA (Medical Software System Developer)

AWARDS & EXHIBITIONS

- **3rd Place (MENA) & Top 50 (Globally), Google Solution Challenge (2022):** Recognized out of thousands of global entries by Google for developing high-impact, scalable technology solutions.
- **1st Place Winner, PSED Challenge:** Awarded first place for the engineering, integration, and presentation of the Dr. HEMA, EV-ion, and Dr. DERMA robotics and software systems.
- **Exhibitor, Annual Science Fair of the Military Technical College:** Selected to present the Dr. HEMA robot, highlighting its AI-driven healthcare capabilities and >90% diagnostic accuracy.

PUBLICATIONS

Elshafee, A. *“HEMA: A Proposed Robot for Improving Healthcare Access in Underserved Communities”*

Published on ResearchGate | [View Publication](#)

TECHNICAL SKILLS

- **Programming & Software:** Python, C, C++, Kotlin, Flutter, Flask, MATLAB, SolidWorks
- **Embedded Systems & Hardware:** ESP32, ESP8266, Arduino, I2C, Serial Communication, Sensor Integration
- **Core Competencies:** Machine Learning, Computer Vision, Biomedical Signal Processing, IoT Systems, Cloud Computing (GCP)

CERTIFICATIONS

- **Android Kotlin Developer Certified** | *Google Africa Developer Training Program*
- **TensorFlow Developer Certificate** | *TensorFlow* (Mar 2022) | Credential ID: 49151162
- **Google Project Management Professional Certificate** | *Coursera* (Mar 2022)
 - *Includes: Project Initiation & Successful Project Delivery*
- **DeepLearning.AI TensorFlow Developer Specialization** | *Coursera* (Jan 2022) | Credential ID: QKRRX657NWZ3
 - *Includes: Convolutional Neural Networks (CNNs), Natural Language Processing (NLP), Sequences, Time Series and Prediction*
- **Flutter Development Training** | *Flutter Philippines* (Mar 2021) | Credential ID: 45689660141581
- **Flutter & Dart - The Complete Guide** | *Udemy* (Mar 2022) | Credential ID: UC-4789a888-22e9-4ec4-9a24-c00ce691e70d
- **Android Development Track & Android Basics** | *Udacity* (2019 – 2020)
- **English A2 Level** | *British Council* (Jan 2021)

LEADERSHIP & VOLUNTEERING

- **Mentor**, Google Developer Groups (Aug 2022 – May 2024)
- **Assistant Head**, IEEE Horus University (Nov 2021 – Oct 2023)
- **Founder & Leader**, GDSC Horus University (Jul 2021 – Aug 2022)
- **Ambassador**, Huawei ICT Horus University (Aug 2021 – Jul 2022)